Overview: MNLI, every split by premise

One premise, one hypothesis per label

Premise: *You outwitted me.*

- **Entailment (E)** — *I was outwitted.*
- **Neutral (N)** — *It was the first time I've been outwitted.*
- **Contradiction (C)** — *You have never outwitted me.*

Splits: no premise on two sides

Tuning 10% — **CV pool** 90%, 5 folds, repeated 3 times

	fold 1	fold 2	fold 3	fold 4	fold 5

39,359 pairs: settings, early stopping — 353,343 pairs: all comparisons

Matched 9,815: same genres (fiction, government, slate, telephone, travel)

Mismatched 9,832: new genres (9/11, face-to-face, letters, OUP, verbatim)

Official validation sets: final report only

Williams, Nangia & Bowman (NAACL 2018). Test labels are not public.

Train corpus

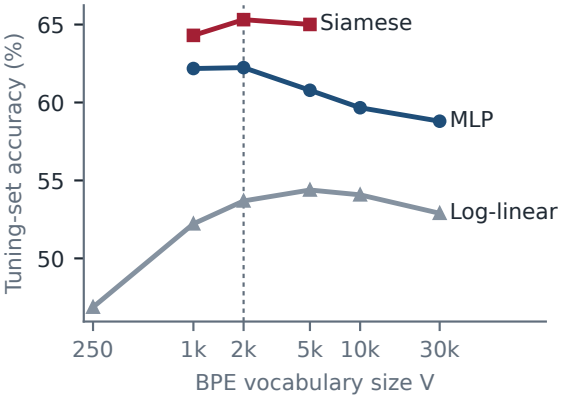
Pairs	392,702
Premises (`promptID`)	128,107
Mean pairs per premise	3.07
Pairs per label	33.3% each
Mean words, premise	19.8
Mean words, hypothesis	10.0

Choices

Tokeniser	BPE, $V = 2000$
Features	Indicator, BoW, BoW + overlap
Models	Log-linear, MLP, Siamese MLP
Metrics	Accuracy, NLL
Compare	Bayesian correlated t, ROPE ± 1 pp

Q2 Tokeniser: BPE, $V = 2000$, trained on our own pool

Vocabulary size (one fit per point)



Training data: is MNLI large enough?

Unique premises + hypotheses	115,349 + 353,343
Running words / word types	5.8M / 179k
`[UNK]` on validation sets	$\leq 0.001\%$
External corpus	none
Tuning, validation sets	excluded

Other settings (log-linear, BoW)

Change from case-sensitive BPE	Δ accuracy
Refit inside each fold (CV)	-0.03 pp
Lowercase, $V = 1$k / 10k	-0.04 / -0.07 pp
Word-level, $V = 1$k / 10k	$+0.04$ / $+0.26$ pp
Train / tuning acc., $V = 2$k	54.7 / 53.7%
Train / tuning acc., $V = 30$k	63.4 / 52.9%

What BPE produces

`_You`	`_have`	`_never`	`_out`	`w`	`itt`	`ed`	`_me.`

No spread to draw: settings were not repeated. Tuning-set SE ≈ 0.25 pp, about the size of a marker. Differences on the tuning set unless marked CV.

Q3 Features: keep the sentences apart and encode what they share

The three representations (one input per pair)

Name	Input	Dims
Indicator	$[\mathrm{ind}(p);\ \mathrm{ind}(h)]$	4,000
BoW	$[\mathrm{cnt}(p);\ \mathrm{cnt}(h)]$	4,000
BoW + overlap	$[\mathrm{cnt}(p);\ \mathrm{cnt}(h);\ \mathrm{ind}(p) \odot \mathrm{ind}(h)]$	6,000

ind: present, 0/1. cnt: count. Overlap: AND of indicators, shaded below.

p	_You	_out	w	itt	ed	_me.	
h	_You	_have	_never	_out	w	itt	ed	_me.

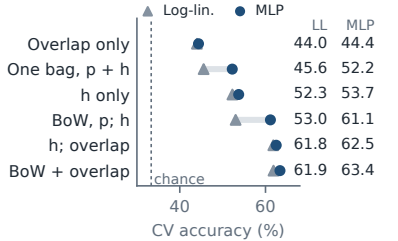
Weights for contradiction (log-linear, mean of 5 folds)

Token	Hypothesis block	Overlap block	Hyp. + overlap
_never	+2.10	−2.47	−0.37
_no	+1.77	−1.77	0.00
_not	+0.89	−1.13	−0.24

Shared tokens per pair (mean count): E 10.3, N 7.4, C 7.6; SD 5–7

E *You've never had a job.* → *You've never been employed.*

What each input gives (mean of 15 fits)



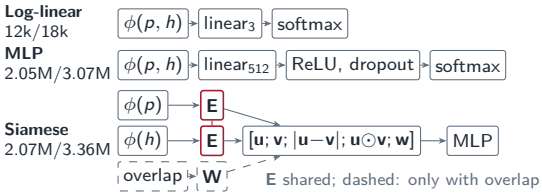
Alternatives (tuning accuracy, %)

	Log-lin.	MLP
BoW	53.7	62.0
TF-IDF per sentence	53.9	63.3
Hashed cross-unigrams	51.7	59.0
BoW + overlap	**62.5**	**64.4**

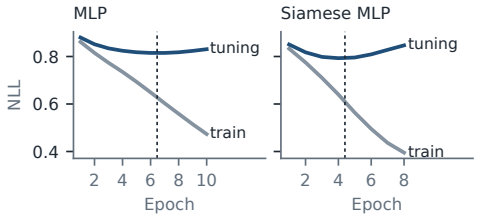
Weight SD over folds ≤ 0.03; the sum leaves out the premise block. Plot SD ≤ 0.28 pp; LL: log-linear. Alternatives: one fit each. Cross-unigrams: Bowman et al. (EMNLP 2015).

Q4 Architectures: from additive scores to an explicit comparison

Three models (parameters: BoW / BoW + overlap)



Train and tuning loss (overlap, mean of 15 fits)



Search on the tuning set

Setting	Chosen	Alternative: Δ pp
Learning rate, AdamW	3×10^{-4}	10^{-4}: −0.1
Weight decay, AdamW	0.3	0.1: −0.1
Hidden units	512	128: −0.6
Dropout	0.3	0.5: +0.1
Activation	ReLU	GELU: −0.6
Hidden layers	1	2: −1.2
Log-linear C	1	flat from 0.1

Mean Δ over 6 neural systems, one fit each (layers: 3 MLPs). tanh: −3.7 pp. SE ≈ 0.25 pp: dropout 0.5 is within noise, −0.1 on the best system.

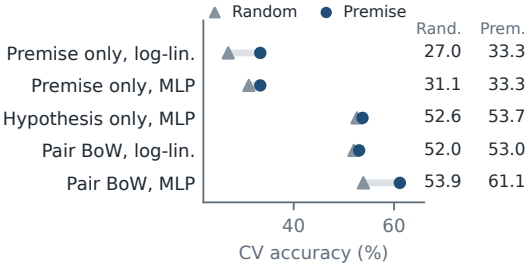
First guess against chosen setting

Tuning, %	lr 10^{-3}, wd 10^{-4}	lr 3×10^{-4}, wd 0.3
MLP, BoW	60.5	**62.1**
Siamese, BoW	65.0	**65.2**
Siamese, overlap	65.5	**66.3**

Conneau et al. (EMNLP 2017). AdamW, batch 256, ≤20 epochs, patience 3; log-linear: L-BFGS. Dashed: mean selected epoch. Train loss on 20k pairs, no dropout.

Q5 Evaluation: the tutorial's random split would mislead us here

Random against premise split (5-fold mean, seed 0)



Held-out accuracy, % (Siamese + overlap)

	Matched	Mismatched
Single model (mean of 15)	64.4	65.0
Majority vote of 15	**65.8**	**66.3**
Hypothesis-only MLP, vote	54.0	53.3
CBOW (published, test)	64.8	64.5
ESIM (published, test)	72.3	72.1
Hypothesis only (published, test)	53.9	52.3

Where it fails (vote)

	Matched	Mismatched
Recall E / N / C	69.4 / 64.7 / 63.1	71.5 / 63.6 / 63.5
Lowest genre	slate 61.2	verbatim 64.4
Highest genre	government 69.0	letters 70.2

Hypothesis only (CV, 53.7 against 64.9): 65% of the best gain over chance

Other checks

MLP − log-linear, BoW: random / premise	1.9 / 8.1 pp
Max |macro-F1 − accuracy| in the grid	0.26 pp
Early stopping on fold, not tuning set	−0.01 pp

Williams et al. (NAACL 2018); Gururangan et al. (NAACL 2018). Binomial SE: held-out 0.5, per label 0.8, per genre 1.1 pp. SD: 15 single models $\leq$ 0.45, folds in the plot $\leq$ 0.7 pp. Selection frozen first.

Q6 Decision: Siamese MLP; the overlap block is not yet decided

CV accuracy, % (mean of 15 fits)

	Log-lin.	MLP	Siam.
Indicator	53.1	61.3	64.0
BoW	53.0	61.1	63.9
BoW + overlap	61.9	63.4	**64.9**

CV NLL (mean of 15 fits)

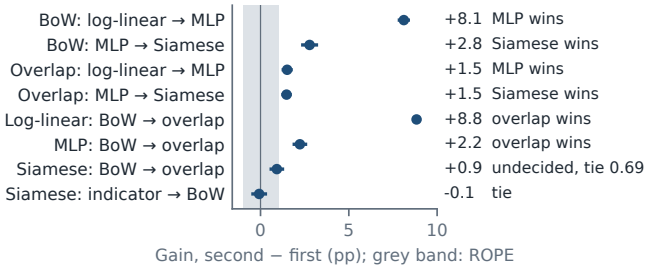
	Log-lin.	MLP	Siam.
Indicator	.972	.860	.816
BoW	.973	.864	.820
BoW + overlap	.862	.826	**.800**

SD over fits: 0.16–0.32 pp; NLL .002–.003

Why ROPE ±1 pp

Held-out SE (9.8k pairs)	0.5 pp
SD between CV repetitions	$\leq$ 0.12 pp
Benavoli et al.	1%

Paired gain in CV accuracy: posterior mean and 95% interval



All 36 pairs at ±1 pp

27 decided, 7 ties, 2 undecided. At ±0.5 pp 6 change, at ±2 pp 8 change, none reverses.

Compare with the posterior, not the SDs: fits share data. 15 differences, paired by repetition and fold; each repetition has new folds and a new initialisation. Benavoli et al. (JMLR 2017): correlated t, $\rho = 1/5$, `runs=3`. Decided: posterior probability above 0.95.